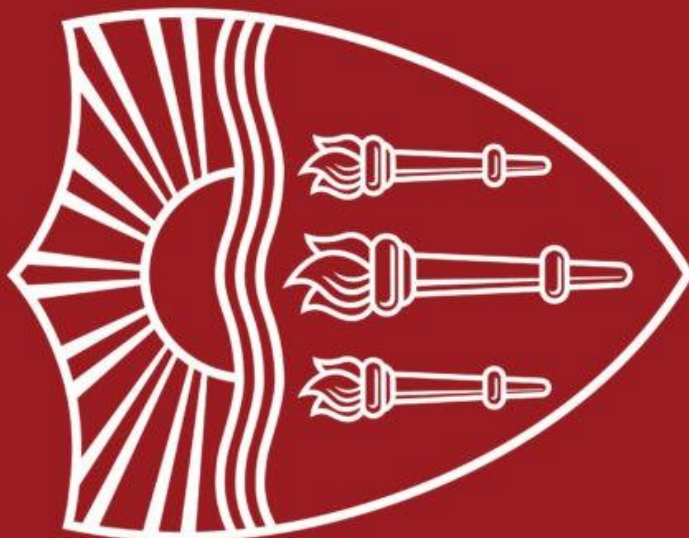




Lecture 6: Word Embeddings (contd.)

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USC CSCI 544 Applied NLP
Fall 2026



Announcements + Logistics

Project Proposal due today EOD

HW1 is on preparation & testing

Lecture Outline

- Announcements
- Recap: Multinomial LR
- Word Embedding
- word2vec
- GloVe
- Properties and Evaluation of Word Embeddings

Recap: Multinomial LR

Multinomial Logistic Regression

The probability of everything must still sum to 1

$K > 2$

$$P(y_1|x) + P(y_2|x) + \cdots + P(y_K|x) = 1$$

- Introducing the softmax function, which
 - Takes a vector $\mathbf{Z} = [z_1, z_2, \dots, z_K]$ of K arbitrary values
 - each z_i corresponds to weighted sum of features for the i -th class
 - Outputs a probability distribution
 - each value in the range $[0,1]$
 - all the values summing to 1

Softmax

The Softmax Function

Turns a vector $\mathbf{z} = [z_1, z_2, \dots, z_K]$ of K arbitrary values into probabilities

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)} \quad 1 \leq i \leq K$$

The denominator $\sum_{j=1}^K \exp(z_j)$ used to normalize all the values into probabilities.

$$\text{softmax}(\mathbf{z}) = \left[\frac{\exp(z_1)}{\sum_{j=1}^K \exp(z_j)}, \frac{\exp(z_2)}{\sum_{j=1}^K \exp(z_j)}, \dots, \frac{\exp(z_K)}{\sum_{j=1}^K \exp(z_j)} \right]$$

Word Embeddings

Case Study: Word Neighborhood Predictor

- Yet do I fear thy nature. It is too full o'the milk of human kindness To catch the nearest way.
- If it were done when 'tis done, then 'twere well It were done quickly.
- Is this a dagger which I see before me, The handle toward my hand?
- My hands are of your color, but I shame To wear a heart so white.
- Knock, knock, knock! Who's there,

Where is it more likely to find the word "doth"?



Case Study: Word Neighborhood Predictor

What words are likely to co-occur with the word “garnish”?



Ingredients

12 Large Eggs, Hard-boiled And Peeled
(See Note)

1/2 c. Mayonnaise

2 tbsp. Prepared Yellow Mustard

1/4 tsp. Kosher Salt

[See Nutritional Information](#) ▼

Directions

Slice eggs in half lengthwise. Pop out yolks and place them in a medium-sized mixing bowl. Mash yolks with a fork and then add mayonnaise, mustard, and salt. Mix until smooth. To achieve maximum smoothness, blitz the yolk mixture a few times with an immersion blender. If you are planning to use a pastry bag and piping tip to add the yolk mixture to the egg whites, it helps to get the yolk mixture nice and smooth.

Divide the yolk mixture evenly between the egg white halves. Garnish as desired. Serve immediately or refrigerate until ready to serve.

Word: Types & Tokens

“The cat chased the cat.”

Word types: *the, cat, chased* → **3 types**

Word tokens: *the, cat, chased, the, cat* → **5 tokens**

Word: Types & Tokens

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Word tokens: *the, cat, chased, the, cat* → **5 tokens**

So:

Word = the linguistic item

Type = the category representing identical items

What do words mean?

A **sense** or “concept” is the meaning component of a word

- Can be polysemous (have multiple senses)

Lemmas

- Canonical form
- For example,
break, breaks, broke, broken and breaking all share the lemma “break”
- Different from “stem”

Dictionary

Definitions from [Oxford Languages](#) · [Learn more](#)



ob·jec·tive

/əbˈjektɪv/

Lemma

adjective

1. (of a person or their judgment) not influenced by personal feelings or opinions in considering and representing facts.

"historians try to be objective and impartial"

Similar:

impartial

unbiased

unprejudiced

nonpartisan

disinterested



2. **GRAMMAR**

relating to or denoting a case of nouns and pronouns used as the object of a transitive verb or a preposition.

noun

1. a thing aimed at or sought; a goal.

"the system has achieved its objective"

Similar:

aim

intention

purpose

target

goal

intent

object

end



2. **GRAMMAR**

the objective case.

Sense

Relation: Synonymy

Synonyms: words that have the same meaning in some or all contexts

- couch / sofa
- big / large
- automobile / car
- water / H₂O
- vomit / throw up

Perfect Synonymy might not be possible...

- Even if many aspects of meaning are identical
- Still may differ based on politeness, slang, register, genre, etc.
- e.g. cannot use H₂O in a surfing guide!
- e.g. my big sister != my large sister

Relation: Similarity

Words with similar meanings. Not synonyms, but sharing some element of meaning

- car, bicycle
- cow, horse

word1	word2
vanish	disappear
behave	obey
belief	impression
muscle	bone
modest	flexible
hole	agreement

Human assessment of
word similarity

SimLex-999 dataset
(Hill et al., 2015)

Relation: Antonymy

Senses that are opposites with respect to only one feature of meaning

Otherwise, they are very similar!

- e.g. dark/light, short/long, fast/slow, rise/fall, hot/cold, up/down, in/out

More formally: antonyms can

- define a binary opposition or be at opposite ends of a scale
 - e.g. long/short, fast/slow
- be reversives: denote opposing processes
 - rise/fall, up/down

Relation: Others

Relatedness

- coffee, tea: similar
- coffee, cup: Related, not similar

Connotation

- Words have affective meanings
 - Positive connotations (happy)
 - Negative connotations (sad)

WordNet

WordNet® is a large lexical database of English

Nouns, verbs, adjectives and adverbs are grouped into sets of synonyms (synsets), each expressing a distinct concept

Relations between synsets:

- Super-subordinate relations (hyperonymy, hyponymy or ISA relation)

 - an armchair is a kind of chair, chair is a kind of furniture

- Meronymy (part-of)

 - chair has legs

- Antonymy



n-grams and Semantics

- n-grams do not represent meaning well
 - Do not tell us that the word “rancor” is close in meaning to the word “hatred”
 - Or that “Rise” and “Fall” have opposite meanings
 - Let alone more complex is-a or part-of relations
- Discrete representations of meaning
- Later: feature representations which are continuous

Computational Models of Word Meanings

Can we build a theory of how to represent word meaning, that accounts for at least some of the desiderata?

We will introduce vector semantics

- The standard model in language processing
- Handles many of our goals

“You shall know a word by the company it keeps.”

- Firth (1957)

What does recent English borrowing ongchoi mean?

Suppose you see these sentences:

- Ongchoi is delicious sautéed with garlic.
- Ongchoi is superb over rice
- Ongchoi leaves with salty sauces

And you've also seen theses:

- ... spinach sautéed with garlic over rice
- Chard stems and leaves are delicious
- Collard greens and other salty leafy greens

Conclusion: Ongchoi is a leafy green like spinach, chard, or collard greens

- We could conclude this based on words like 'leaves', 'delicious', and 'sautéed'

Ongchoi: “Water Spinach”

空心菜

Kangkung

Rau muống

...



Yamaguchi, Wikimedia Commons, public domain

Idea 1: Define meaning by linguistic distribution

Let's define the meaning of a word by its distribution in language use, meaning its neighboring words or grammatical environments.

Idea 2: Meaning as a point in space (Osgood et al. 1957)

3 affective dimensions for a word

- Valence: pleasantness
- Arousal: intensity of emotion
- Dominance: the degree of control exerted

	Word	Score		Word	Score
Valence	love	1.000		toxic	0.008
	happy	1.000		nightmare	0.005
Arousal	elated	0.960		mellow	0.069
	frenzy	0.965		napping	0.046
Dominance	powerful	0.991		weak	0.045
	leadership	0.983		empty	0.081

NRC VAD Lexicon
(Mohammad 2018)

Hence the connotation of a word is a vector in 3-space

Idea 1: Define meaning by linguistic distribution

Idea 2: Meaning as a point in multidimensional space

Defining meaning as a point in space based on distribution

Defining meaning as a point in space based on distribution

- Each word = a vector
- not just “good” or “word#45”
- Similar words are “nearby in semantic space”
- We build this space automatically by seeing which words are nearby in text
- 2-D representation



From now on:

Computing with meaning representations instead of string representations

荃者所以在鱼，得鱼而忘荃
言者所以在意，得意而忘言

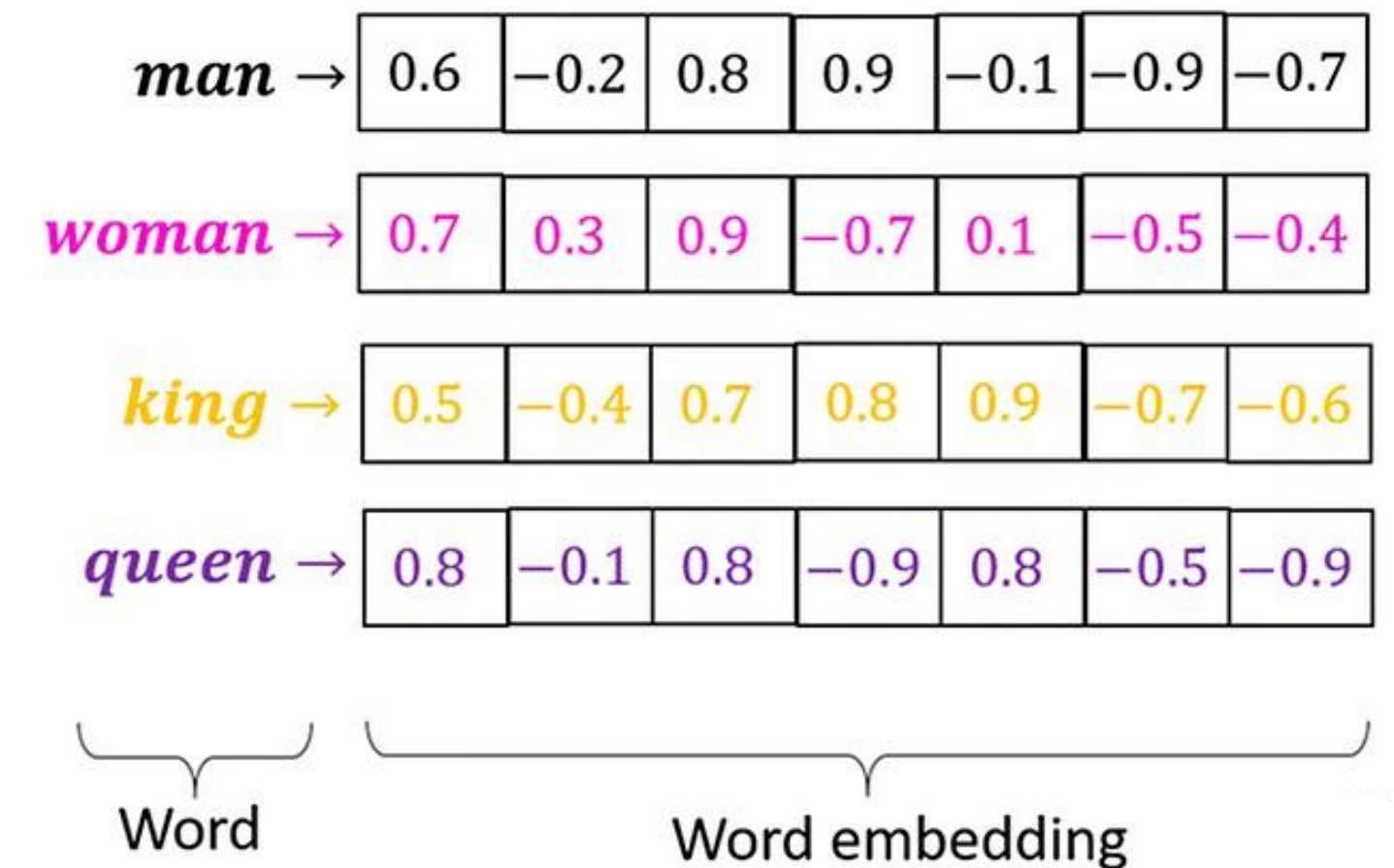
— 庄子 (369 BC - 286 BC)

Nets are for fish;
Once you get the fish, you can forget the net.

Words are for meaning;
Once you get the meaning, you can forget the words.

Words as Vectors

- In NLP, we commonly represent word types with vectors!
- Very useful in capturing similarity between words, and other forms of lexical semantics (e.g. synonymy, hypernyms, antonymy)
- Computing the similarity between two words (or phrases, or documents) is extremely useful for many NLP tasks
 - Q: How tall is Mount Everest?
 - A: The official height of Mount Everest is 29029 ft



- Similarity for plagiarism detection
- Word similarity can lead to sentence and document similarity

enough scale for companies to make profit from it. In order to be competitive with new technologies, the challenge of today's large companies is to create new business within their business (Garvin & Levesque, 2006). Furthermore, the two researchers emphasize a switch from downsizing and cost cutting to the creation, development and assistance of innovative new businesses. For existing companies the implementation of corporate entrepreneurship, in order to develop innovative businesses, is risky. Are the three types of entrepreneurship linked together over time? How long does it take to change behavior of the firm as a whole? If the five attributes are created, do all grow together equally, or do some grow faster and earlier than others? How do the importance and intensities of the attributes differ both absolutely and relatively in each type? These are the questions that a longitudinal study such as this can attempt to answer to shed light on the nature of organizations' adjustments to hostile environments. According to Garvin and Levesque (2006) implementing new ventures face several barriers, and can only be successful if a blend of old and new organizational traits is done. To achieve a blend of old and new, an organization needs to rely on employee innovative behavior in order to succeed in dynamic business environments (Yuan & Woodman, 2010).



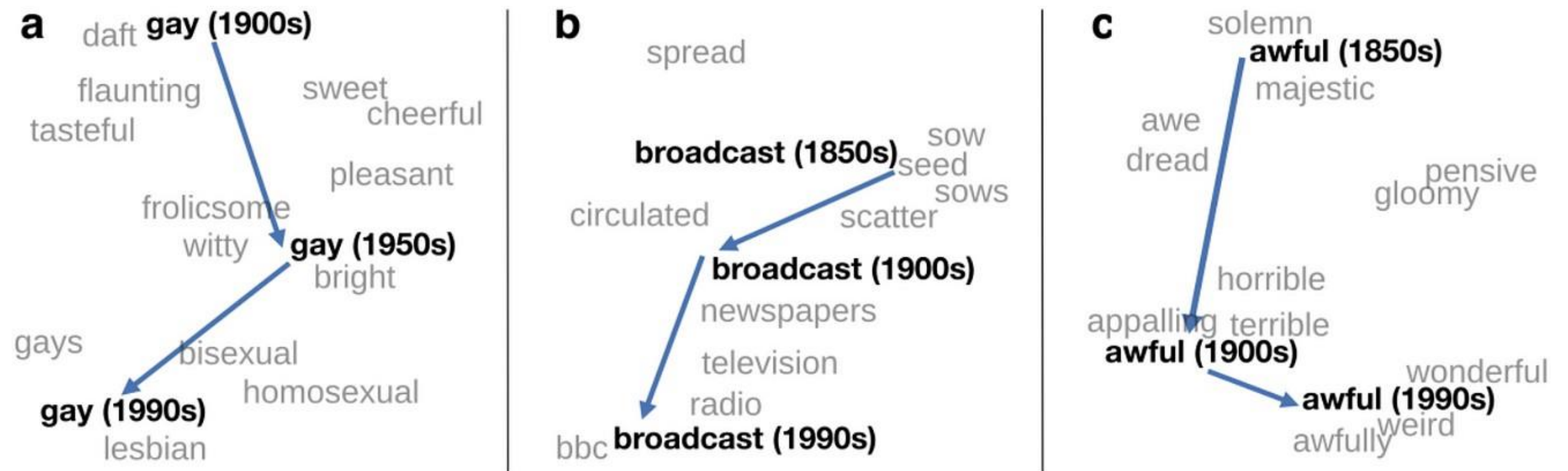
Figure 1 - Corporate entrepreneurship model - literature review.com

The downside is potentially very damaging to a startup's lifespan: if a startup lands a pilot or POC with the corporation running the accelerator, they have very little bargaining power or time to find other partners to test their solution with. The transition from manufacturing economy to service economy has led to a shift in business agenda from

1 Original source
onlinelibrary.wiley.com/stor...

...all grow together equally, or do some grow faster and earlier than others? These are the questions that a longitudinal study such as this can attempt to answer to shed light on the nature of organizations' adjustments to hostile environments. Of the many ways to adjust, two stand out at

- Visualizing semantic change over time
- New words: dank, cheugy, rizz, shook, situationship, simp, delulu, cottagecore



~30 million books, 1850-1990, Google Books data

Discrete representations

So far, words are regarded as atomic symbols: hotel, conference, walk

Missed nuances: synonyms, antonyms

WordNet taxonomy

Words as Vectors

“You shall know a word by the company it keeps.”

- Firth (1957)

Word Meaning via Language Use

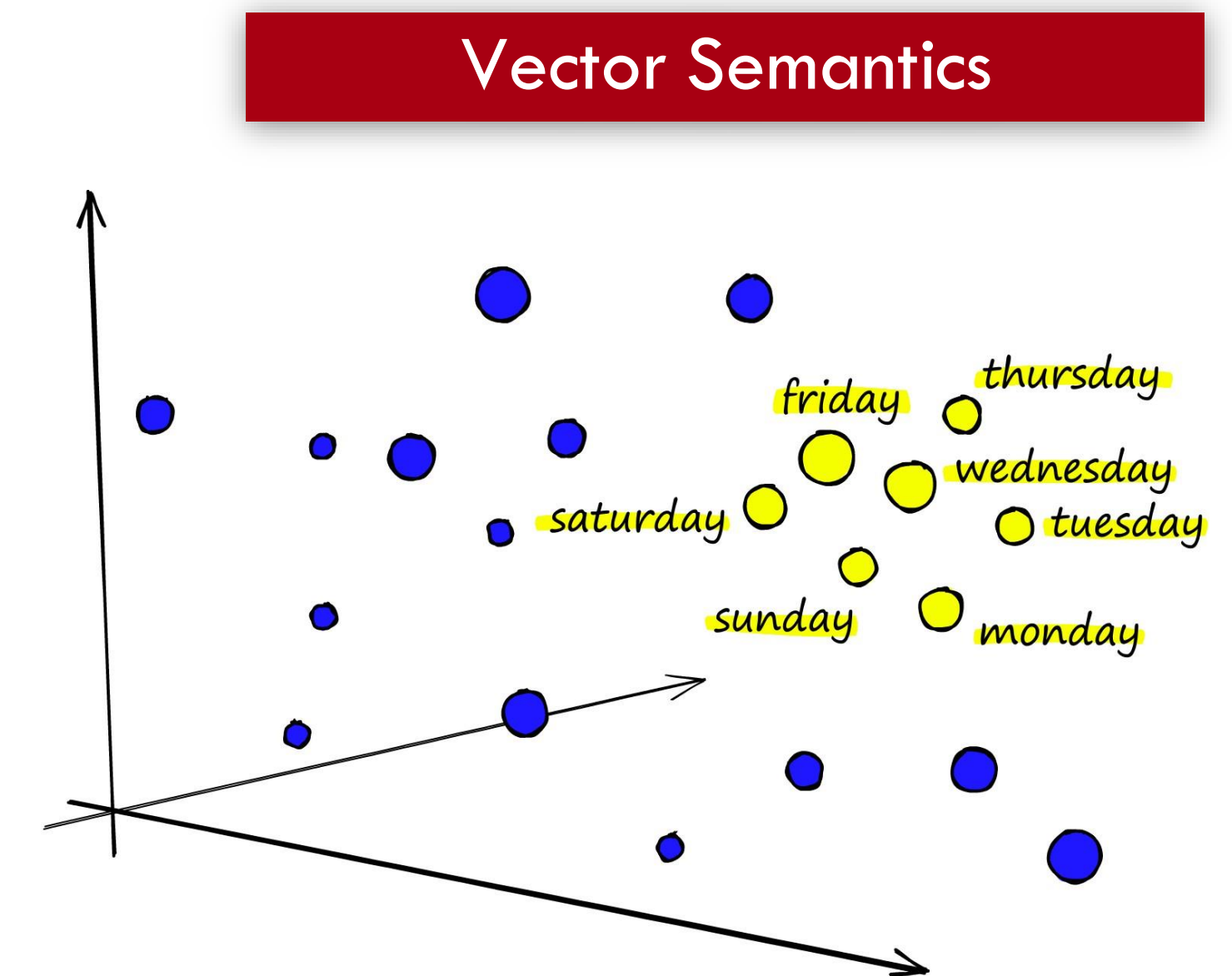
- The meaning of a word can be given by its distribution in language usage
 - Word “usage”: word environments / contexts
 - Neighboring words or grammatical environments
- Intuitions: Zellig Harris (1954):
 - “oculist and eye-doctor ... occur in almost the same environments”
 - “If A and B have almost identical environments we say that they are synonyms.”
- A bottle of tesgüino is on the table
- Everybody likes tesgüino
- Tesgüino makes you drunk
- We make tesgüino out of corn.



Two words are similar if they have similar word contexts

Word Embeddings

- Represent a word as a point in a multidimensional semantic space
 - Space itself constructed from distribution of word neighbors
- Called an "embedding" because it's embedded into a space
- Fine-grained model of meaning for similarity



Every modern NLP algorithm uses embeddings as the representation of word meaning

Intuition: Why Vectors?

Consider generation:

- With word strings, a feature is a word identity
- Feature 5: 'The previous word was "horror"'
- Requires exact same word to be in training and test

With embeddings:

- Feature is a word vector
- 'The previous word was vector [35,22,17...]'
- Now in the test set we might see a similar vector [34,21,14] ... perhaps corresponding to "horrific"
- We can generalize to similar but unseen words!!!

Cosine Similarity for Word Similarity

Cosine similarity of two vectors

$$\cos(\vec{v}, \vec{w}) = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}| |\vec{w}|} = \frac{\sum_{i=1}^N v_i w_i}{\sqrt{\sum_{i=1}^N v_i^2} \sqrt{\sum_{i=1}^N w_i^2}}$$

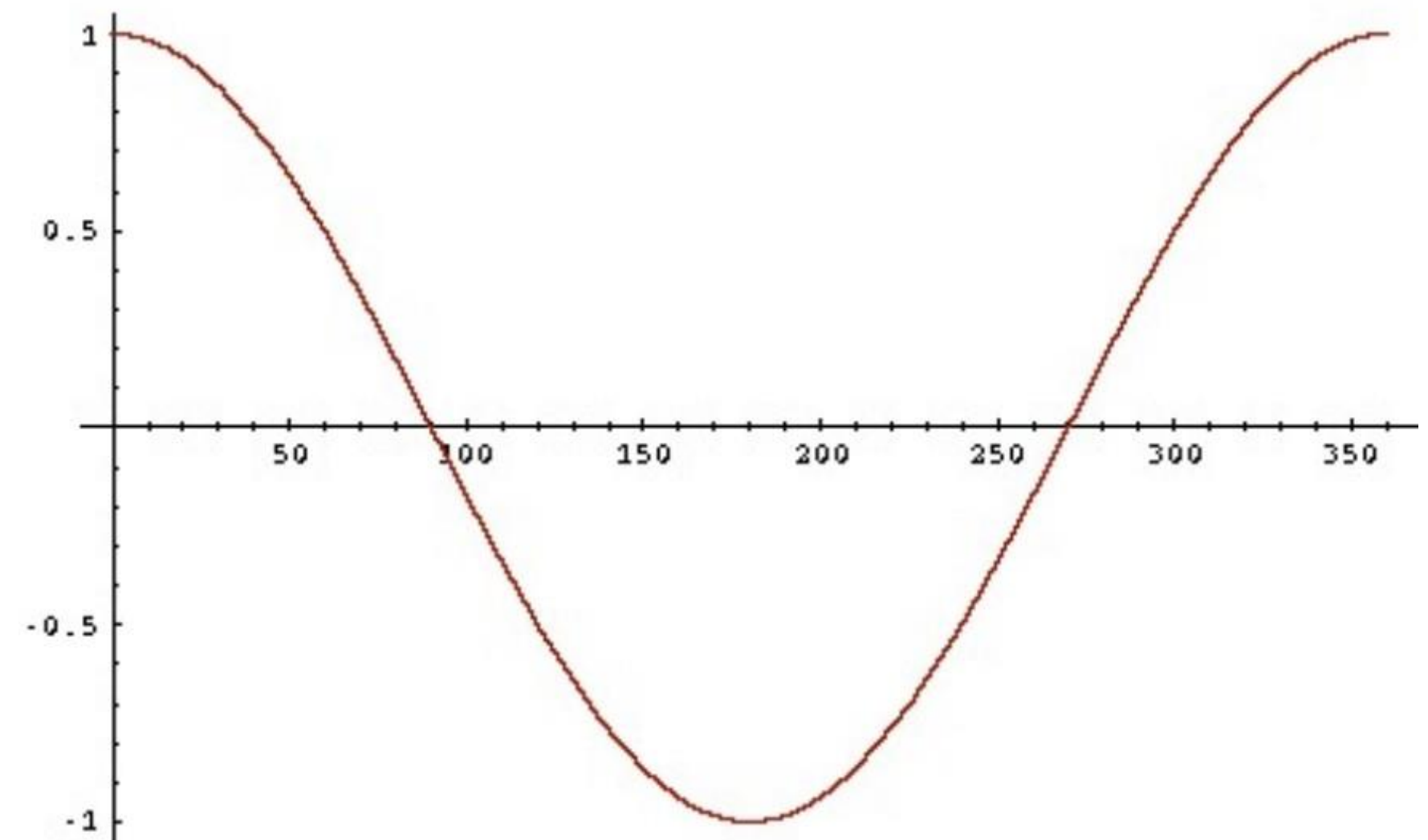
Based on the definition of the dot product
between two vectors $\vec{a}$ and $\vec{b}$

$$\vec{v} \cdot \vec{w} = |\vec{v}| |\vec{w}| \cos \theta$$

$$\cos \theta = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}| |\vec{w}|}$$

Cosine as a similarity metric

-1: vectors point in opposite directions
+1: vectors point in same directions
0: vectors are orthogonal



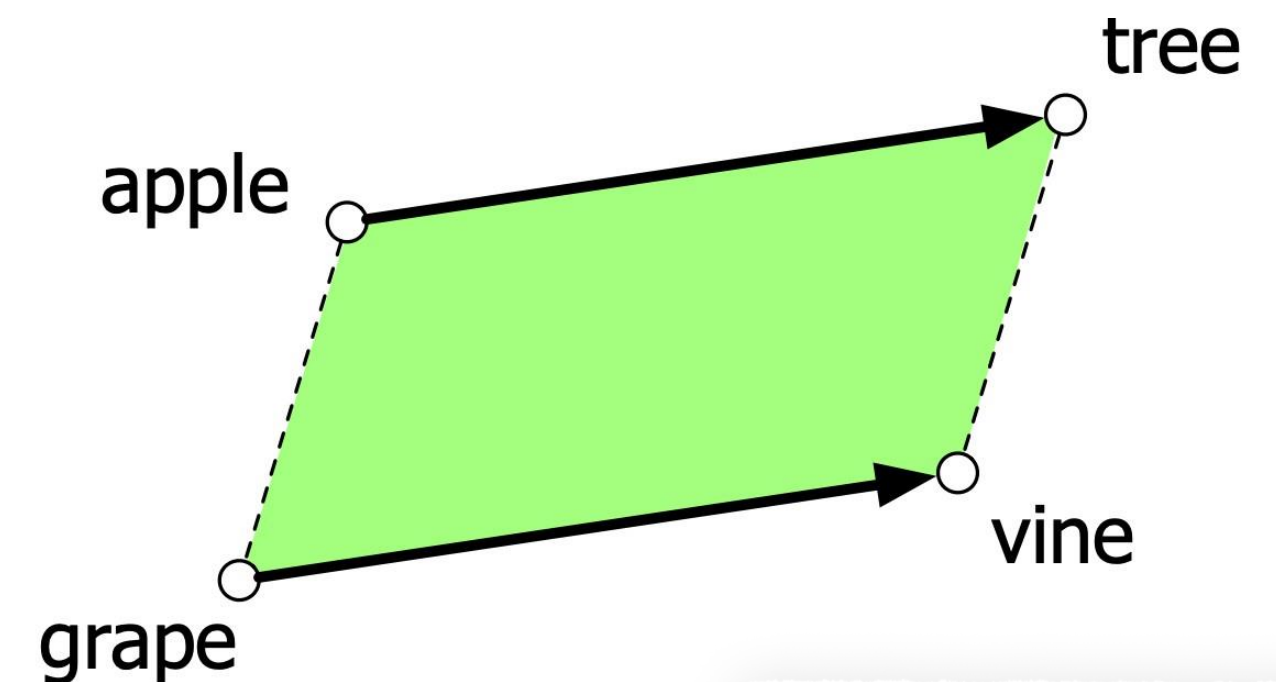
Greater the cosine, more similar the words

Analogy Relations

- The classic parallelogram model of analogical reasoning
 - Word analogy problem:
 - “Apple is to tree as grape is to ...”

Maximize similarity = minimize distance

Add $(\mathbf{w}_{apple} - \mathbf{w}_{tree})$ to $\mathbf{w}_{grape}$...
Should result in $\mathbf{w}_{vine}$



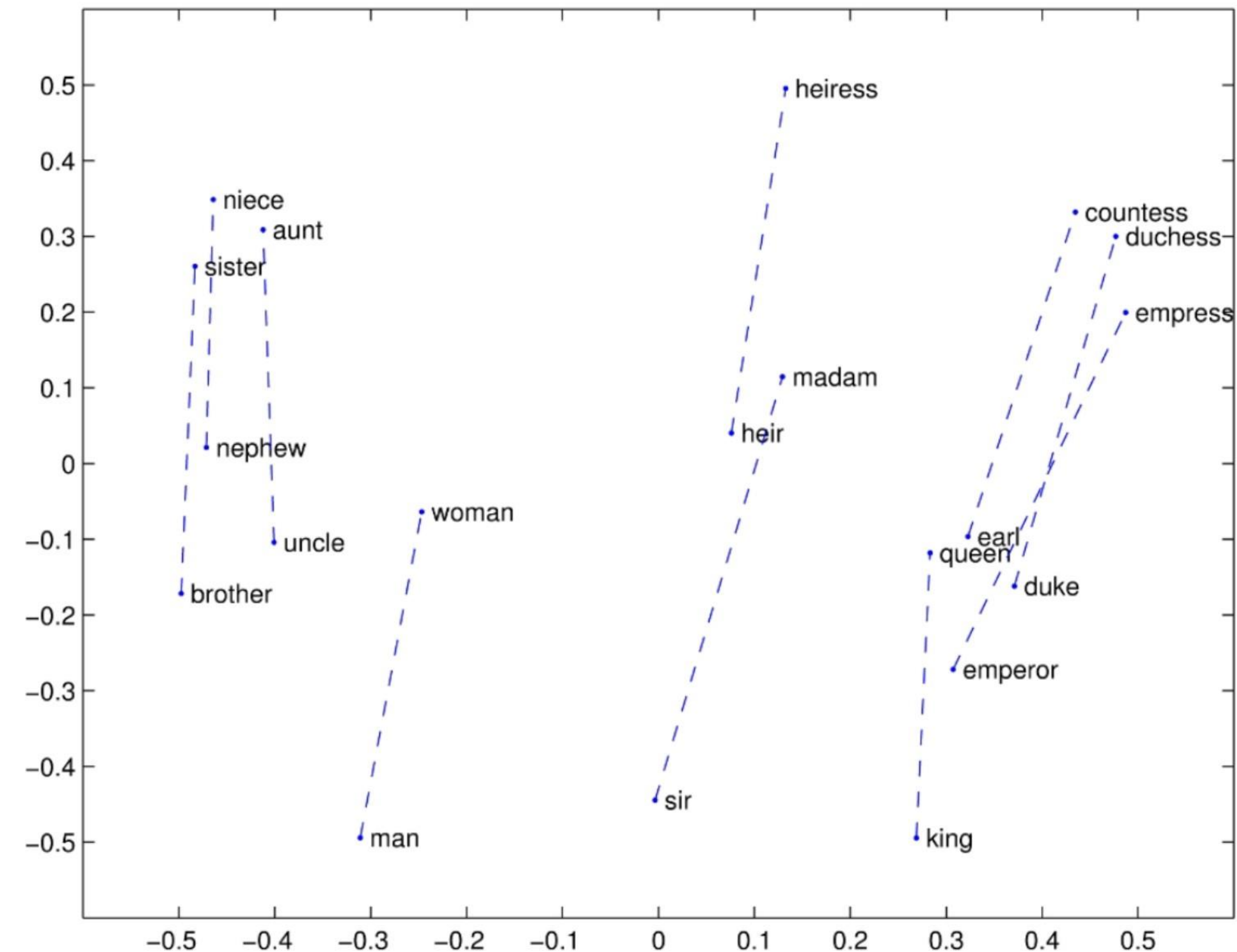
Rumelhart and Abrahamson, 1973

For a problem $a : a^* :: b : b^*$, the parallelogram method is:

$$\hat{b}^* = \arg \max_{\mathbf{w}} \text{sim}(\mathbf{w}, \mathbf{b} - \mathbf{a} + \mathbf{a}^*)$$

Analogy Relations: GloVe Embeddings

- Relational properties of the GloVe vector space, shown by projecting vectors onto two dimensions
- $\mathbf{W}_{king} - \mathbf{W}_{man} + \mathbf{W}_{woman}$ is similar to $\mathbf{W}_{queen}$
- Caveats: Only works for frequent words, small distances and certain relations (relating countries to capitals, or parts of speech), but not others
 - Understanding analogy is an open area of research



Count-based Embeddings

n-grams as One-hot Vectors

vocabulary

i

hate

love

the

movie

film

movie = $\langle 0,0,0,0,1,0 \rangle$

film = $\langle 0,0,0,0,0,1 \rangle$

Unigram Vectors: Represent each word as a vector of zeros with a single 1 identifying the index of the word

One hot vector

How can we compute a vector representation such that the dot product correlates with word similarity?

Dot product is zero! These vectors are orthogonal

Term-document matrix

Let us consider a collection of documents and count how frequently a word (term) appears in each.

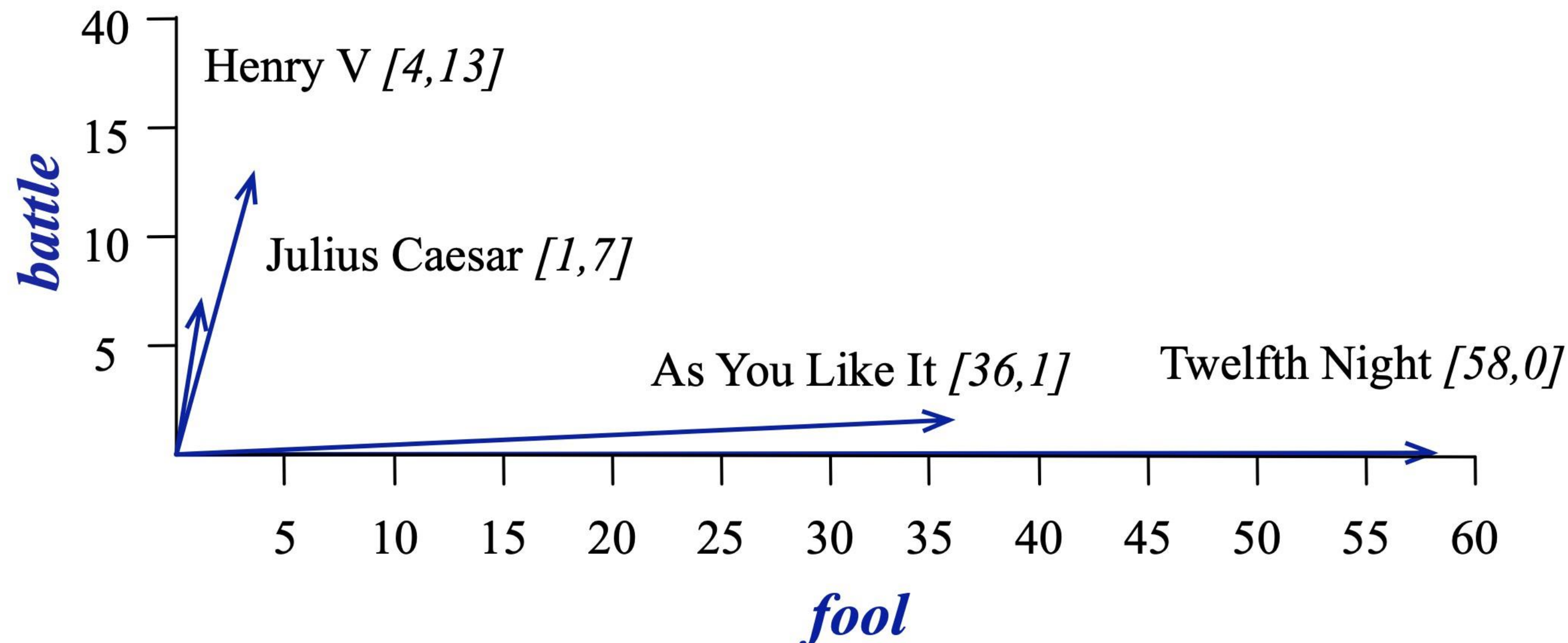
A document could be a play or a Wikipedia article. In general, documents can be anything; we often call each paragraph a document!

Each **document** is represented by a vector of words

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Visualizing document vectors

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3



- Vectors are similar for the two comedies
- Comedies are different from the other two (tragedies)
 - More fools, less battle

Words as vectors in a co-occurrence matrix

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

“Battle” is the kind of word that appears in Julius Caesar and Henry V

“Fool” is the kind of word that appears in As You Like It and Twelfth Night

Number of dimensions?

Words with similar neighborhoods have similar meanings

How to measure a word's **neighborhood**?

Word-context matrix (a kind of co-occurrence matrix)

- each row represents a word in the vocabulary
- each column represents how often each other word in the vocabulary appears

Word-Context Matrix

	aardvark	abacus	adept	affect	agate	...	zydeco
aardvark							
abacus							
adept							
affect							
agate							
...							
zydeco							

How often does
agate occur
near **abacus**?

Word-Context Matrix

What does "nearby" mean?

For right now let's say "within 4 words"

Word-word co-occurrence matrix

is traditionally followed by **cherry** pie, a traditional dessert
 often mixed, such as **strawberry** rhubarb pie. Apple pie
 computer peripherals and personal **digital** assistants. These devices usually
 a computer. This includes **information** available on the internet

	a	computer	pie
cherry	1	0	1
strawberry	0	0	2
digital	0	1	0
information	1	1	0

The 'mini-matrix' for just 4 words and 3 contexts

	a	computer	pie
cherry	1	0	1
strawberry	0	0	2
digital	0	1	0
information	1	1	0

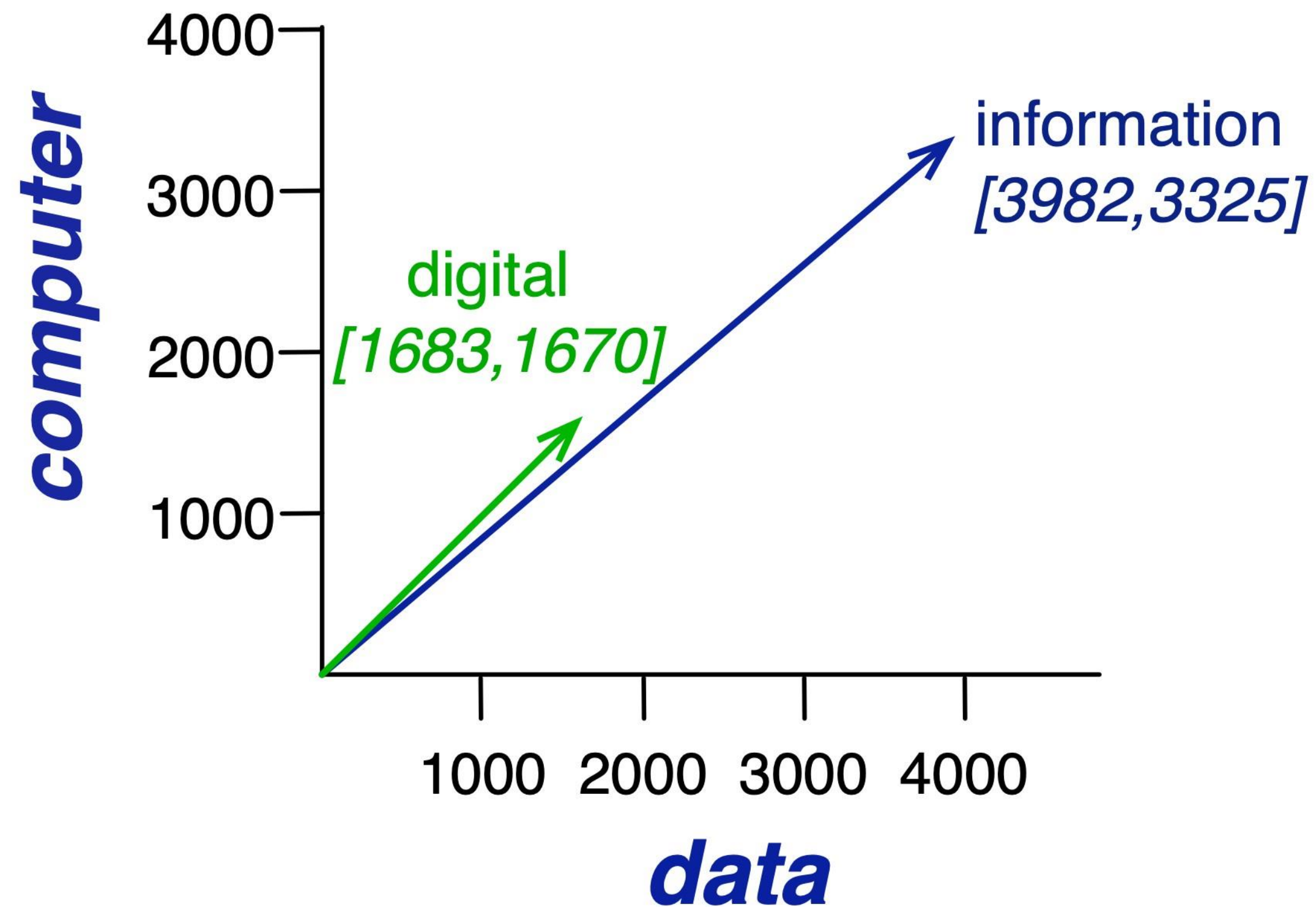
- This 4x3 matrix is a subset of full $|V| \times |V|$ matrix
- Each word is represented by a row vector with dimensionality $[1 \times |V|]$
- With co-occurrence counts with each other word

A selection from a larger word-context matrix

is traditionally followed by **cherry** pie, a traditional dessert
 often mixed, such as **strawberry** rhubarb pie. Apple pie
 computer peripherals and personal **digital** assistants. These devices usually
 a computer. This includes **information** available on the internet

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	...
strawberry	0	...	0	0	1	60	19	...
digital	0	...	1670	1683	85	5	4	...
information	0	...	3325	3982	378	5	13	...

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	...
strawberry	0	...	0	0	1	60	19	...
digital	0	...	1670	1683	85	5	4	...
information	0	...	3325	3982	378	5	13	...



Raw frequencies though...

- ...are a bad representation!
- The co-occurrence matrices we have seen represent each cell by word frequencies
- Frequency is clearly useful; if sugar appears a lot near apricot, that's useful information
- But overly frequent words like *the*, *it*, *or* they are not very informative about the context
- It's a paradox! How can we balance these two conflicting constraints?

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

Need some form of weighting!

Two different kinds of weighting

tf-idf: Term Frequency - Inverse Document Frequency

- Downweighting words like "the" or "if"
- Term-document matrices

PMI: Pointwise Mutual Information

- Considers the probability of words like "good" and "great" co-occurring
- Word co-occurrence matrices

Term Frequency

Term Frequency: frequency counting (usually log transformed)

$$\mathbf{tf}_{t,d} = \begin{cases} 1 + \log(\mathbf{count}(t, d)), & \text{if } \mathbf{count}(t, d) > 0 \\ 0, & \text{otherwise} \end{cases}$$

$\mathbf{count}(t, d)$ = # occurrences of word t in document d

Inverse Document Frequency

Document Frequency: df_t is the number of documents t occurs in.

- NOT collection frequency: total count across all documents
- "Romeo" is very distinctive for one Shakespeare play

- Inverse Document Frequency: idf_t

$$idf_t = \log_{10} \left(\frac{N}{df_t} \right)$$

N = total number of documents in the collection

	Collection Frequency	Document Frequency
Romeo	113	1
action	113	31

Word	df	idf
Romeo	1	1.57
salad	2	1.27
Falstaff	4	0.967
forest	12	0.489
battle	21	0.246
wit	34	0.037
fool	36	0.012
good	37	0
sweet	37	0

What does IDF signify?

tf-idf

Key Uses of TF-IDF [↗](#)

- **Information Retrieval:** Helps to rank documents by relevance in search engines.
- **Document Similarity:** Enables comparison and clustering of similar texts.
- **Text Classification:** Aids in identifying patterns for tasks like spam filtering or sentiment analysis.
- **Keyword Extraction:** Automatically highlights key terms and generates tags or summaries for documents.

$$tf_{t,d} \times idf_{t,d}$$

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Raw Counts

tf-idf Weighted Counts

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.074	0	0.22	0.28
good	0	0	0	0
fool	0.019	0.021	0.0036	0.0083
wit	0.049	0.044	0.018	0.022

Pointwise Mutual Information (PMI)

- PMI between two words:
$$PMI(w_1, w_2) = \log \frac{P(w_1, w_2)}{P(W_1)P(w_2)}$$
 - (association) Do words w_1 and w_2 co-occur more than if they were independent?
- PMI ranges from $-\infty$ to $+\infty$
 - Negative values are problematic: words are co-occurring less than we expect by chance
 - Only reliable under an enormous corpora
 - Imagine w_1 and w_2 whose probability of occurrence is each 10^{-6}
 - Hard to be sure $P(w_1, w_2)$ is significantly different than 10^{-12}
 - So we just replace negative PMI values by 0

- Positive PMI
$$PPMI(w_1, w_2) = \max(0, \log \frac{P(w_1, w_2)}{P(W_1)P(w_2)})$$

Computing PPMI on a term-context matrix

Context c

Term w

	computer	data	result	pie	sugar	count(w)
cherry	2	8	9	442	25	486
strawberry	0	0	1	60	19	80
digital	1670	1683	85	5	4	3447
information	3325	3982	378	5	13	7703
count(context)	4997	5673	473	512	61	11716

Frequency $f(w_i, c_j)$ or f_{ij} is the # times w_i occurs in context c_j

$$P_{ij} = \frac{f_{ij}}{\sum_{t=1}^W \sum_{k=1}^C f_{tk}}$$

$$P_i = \sum_{j=1}^C P_{ij}$$

$$P_j = \sum_{i=1}^W P_{ij}$$

$$PPMI(w_i, c_j) = PPMI_{i,j} = \max(0, \log \frac{P_{ij}}{P_i P_j})$$

	computer	data	result	pie	sugar	count(w)
cherry	2	8	9	442	25	486
strawberry	0	0	1	60	19	80
digital	1670	1683	85	5	4	3447
information	3325	3982	378	5	13	7703
count(context)	4997	5673	473	512	61	11716

$$p(w=\text{information}, c=\text{data}) = 3982/111716 = .3399$$

$$p(w=\text{information}) = 7703/11716 = .6575$$

$$p(c=\text{data}) = 5673/11716 = .4842$$

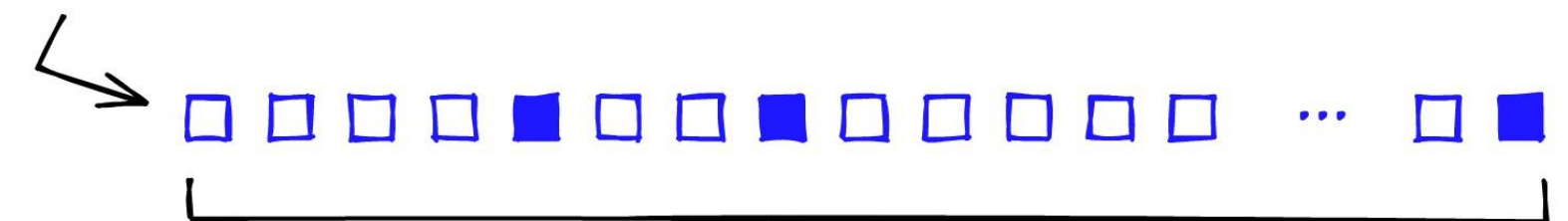
	p(w,context)					p(w)
	computer	data	result	pie	sugar	p(w)
cherry	0.0002	0.0007	0.0008	0.0377	0.0021	0.0415
strawberry	0.0000	0.0000	0.0001	0.0051	0.0016	0.0068
digital	0.1425	0.1436	0.0073	0.0004	0.0003	0.2942
information	0.2838	0.3399	0.0323	0.0004	0.0011	0.6575
p(context)	0.4265	0.4842	0.0404	0.0437	0.0052	

The problem...

- PMI vectors are
 - long (length $|V| = 20,000$ to $50,000$)
 - sparse (most elements are zero)
- Alternative: learn vectors which are
 - short (length 50-1000)
 - dense (most elements are non-zero)

sparse

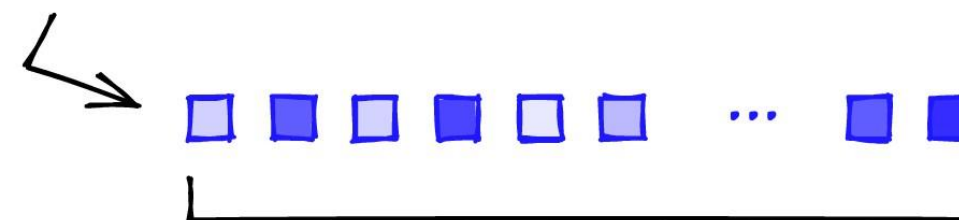
$[0, 0, 0, 1, 0, \dots 0]$



30K+

dense

$[0.2, 0.7, 0.1, 0.8, 0.1, \dots 0.9]$



784

Sparse vs. Dense Vectors

- Why dense vectors?
 - Short vectors may be easier to use as features in machine learning (fewer weights to tune)
 - Dense vectors may generalize better than explicit counts
 - Dense vectors may do better at capturing synonymy:
 - car and automobile are synonyms; but are distinct dimensions in sparse vector
 - a word with car as a neighbor and a word with automobile as a neighbor should be similar, but aren't captured by sparse vector
 - In practice, they work better

How to obtain dense vectors?

- “Neural Language Model”-inspired models
 - Word2vec (skipgram, CBOW), GloVe
- Singular Value Decomposition (SVD)
 - Special case: Latent Semantic Analysis (LSA)
- Alternatives to “static word type embeddings”:
 - Contextual Embeddings (LLM word embeddings)
 - Compute distinct embeddings for a word in its context
 - Separate embeddings for each token of a word

dense

$[0.2, 0.7, 0.1, 0.8, 0.1, \dots 0.9]$

